

Incontinentia Pigmenti

Incontinentia pigmenti (IP), also known as Bloch-Sulzberger syndrome, was first described by Garrod et al. in 1906. It is an X-linked dominant disorder that primarily affects females, as it is typically embryonic lethal in most males. The majority of cases are caused by mutations in the **NEMO** gene (*nuclear factor κ B essential modulator*) located on chromosome **Xq28**.

The condition is characterized by cutaneous lesions that follow the **lines of Blaschko**, progressing through four stages—often with some overlap:

1. **Vesicular stage:** Appears at birth or shortly thereafter.
2. **Verrucous stage:** Develops between 2 and 8 weeks of age.
3. **Hyperpigmented stage:** Emerges in infancy and may persist into adulthood.
4. **Hypopigmentation stage:** Develops from infancy onward, resulting in atrophic, hypopigmented scars.

The initial vesicular stage reflects the presence of **NEMO-deficient cells**. The NEMO protein is essential for activation of the **NF- κ B signaling pathway**, which protects cells from tumor necrosis factor (TNF)-induced apoptosis. In its absence, uncontrolled apoptosis occurs. Over time, these defective cells are eliminated and replaced by cells expressing the normal NEMO allele, leading to resolution of the inflammatory phase.

The **verrucous stage** is thought to result from compensatory hyperproliferation of normal NEMO-expressing keratinocytes. In the **hyperpigmented stage**, melanin is released from damaged basal keratinocytes and transferred to dermal macrophages (melanophages), a process known as **pigment incontinence**. This leads to swirling, hyperpigmented patterns most prominent on the trunk but also present on the limbs.

Histologically, the hyperpigmented areas show: - Increased melanin in the basal layer and dermis - Numerous melanin-laden melanophages - Vacuolization and degeneration of basal keratinocytes

Over time, the hyperpigmentation fades and transitions into the **hypopigmented stage**, characterized by **linear, atrophic, hairless scars** along Blaschko's lines. Histologically, this stage reveals: - Thinned epidermis - Reduced or absent skin appendages - Normal or decreased numbers of melanocytes - Dermal scarring contributing to the hypopigmented appearance

In addition to skin involvement, a significant proportion of patients develop abnormalities in other systems, including: - **Ocular** (e.g., retinal vascular anomalies) - **Dental** (e.g., delayed eruption, peg-shaped teeth) - **Skeletal** (e.g., limb asymmetry, vertebral anomalies) - **Central nervous system** (e.g., seizures, developmental delay)

Linear and Whorled Nevoid Hypermelanosis (LWNH)

Extra-cutaneous abnormalities have been reported in several cases of LWNH, including: - Developmental and growth retardation - Facial and body asymmetry - Ventricular septal defect - Pseudohermaphroditism

Evidence supports that LWNH arises from **genetic mosaicism**, confirmed in some cases by chromosomal analysis showing mosaic trisomy 7, 14, 18, 20, or X-chromosome mosaicism. Diagnosis can be confirmed through chromosomal studies of lymphocytes or cultured dermal fibroblasts.

Histologic findings in LWNH include: - Increased pigmentation of the basal layer - Prominent or vacuolated melanocytes - Pigment incontinence, which is usually absent but may occasionally be present

Congenital Diffuse Reticular Hypermelanosis / Dyskeratosis Congenita

(Note: The provided text ends abruptly. Dyskeratosis congenita is a distinct genodermatosis characterized by the triad of abnormal skin pigmentation, nail dystrophy, and oral leukoplakia, often associated with bone marrow failure. It is not directly related to IP or LWNH.)